

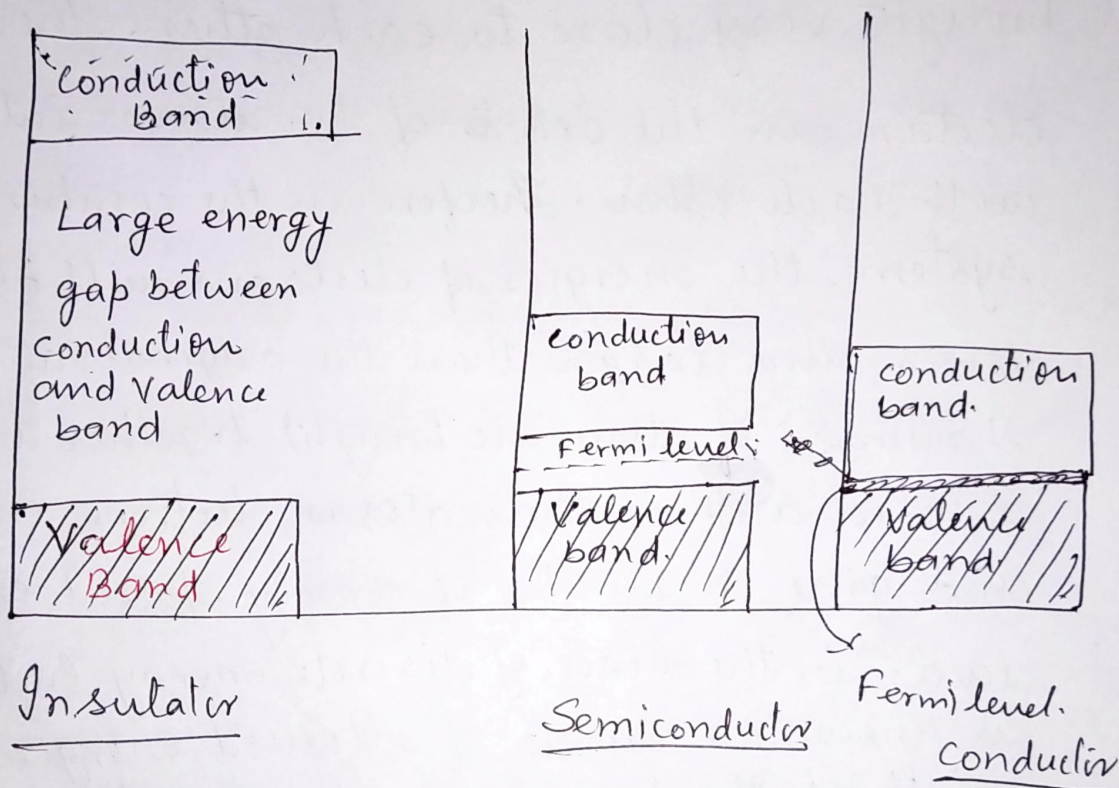
Origin of Energy bands

Since we know that isolated atoms have discrete energy levels. Suppose two isolated atoms are brought very close to each other. Then the electrons in the orbits of two atoms interact with each other. Therefore in the combined system, the energies of electrons will be slightly lower and larger than the original value. If N number of atoms are brought together to form a solid and if these atoms electrons interact and give N number of closely spaced energy levels in the place of discrete energy levels, it is known as bands of allowed energies. The width of the bands depends upon the degree of overlapping of electrons of adjacent atoms and is largest for outermost electrons.

The band corresponding to outermost orbits is called conduction band and next inner band is called valance band. The gap between these two allowed bands is called forbidden energy gap or band gap.

classification of crystalline solids

Based on the width of forbidden gap (band) solids are classified into insulators, semiconductors and conductors.



- (i) conductors $\rightarrow$ The Valence band and conduction bands overlapped with each other and the energy gap E_g is zero. At room temperature free electrons already exists in huge number at conduction band. Hence these solids are good electrical conductors as well as good thermal conductors. The electrical resistivity increases at high temperature by collisions among the free electrons.
- Ex Au, Ag Cu, Al etc.

③ ⑥

Semiconductor $\rightarrow$ The valence band and conduction bands are separated with a small energy band gap $E_g \approx 1.0 \text{ eV}$. At 0 K temperature, free electrons are not available in conduction band and hence they behave like insulators. The electrical conductivity increases at ~~high~~ temperatures by transition of free electrons from valence band to conduction band. Thus these solids behave like conductors at high temperatures.

Ex: Silicon $E_g = 1.1 \text{ eV}$, Germanium $E_g = 0.7 \text{ eV}$

Insulators $\rightarrow$ Valence band and conduction bands are separated by a very large energy gap $E_g > 3 \text{ eV}$. At room temperatures the conduction band is empty and valence band is full of electrons. Hence these solids are electrical insulators. Even at high temperatures valence electrons are unable to jump into the conduction band.

Ex. Glass, Mica, Rubber etc.